Algebra

CHAPTER 4

Principle of Mathematical Induction 4.1–4.2

DPP 4.1 Principle of Mathematical Induction 4.1

Ananth Garg

CHAPTER 4

Principle of Mathematical Induction

DPP 4.1

Principle of Mathematical Induction

Subjective Type Questions

1. Prove that $1 + 5 + 9 + \ldots + (4n - 3) = n(2n - 1)$, for all natural numbers n.

2. Prove that
$$\frac{1}{1+x}+\frac{2}{1+x^2}+\frac{4}{1+x^4}+\ldots+\frac{2^n}{1+x^{2^n}}=\frac{1}{x-1}+\frac{2^{n+1}}{1-x^{2^{n+1}}},$$
where $x \neq \pm 1$ and $n \in N$.

3. Prove that, for all $n \in N$,
$$\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\ldots\left(1-\frac{1}{(n+1)^2}\right)=\frac{n+2}{2n+2}.$$

4. Prove that $2^{3n} - 1$ is divisible by 7, for all natural numbers n.

5. For any natural number n, $7^n - 2^n$ is divisible by 5.

6. Prove that $n^2 < 2^n$ for all natural numbers $n \geq 5$.

7. A sequence $d_1, d_2, d_3, \ldots$ is defined by letting $d_1 = 2$ and $d_k = \frac{d_{k-1}}{k}$, for all natural numbers $k \geq 2$. Show that $d_n = \frac{2}{n!}$, for all $n \in N$.

8. Prove that $\frac{1}{n+1}+\frac{1}{n+2}+\ldots+\frac{1}{2n}>\frac{13}{24}$, for all natural numbers $n > 1$.

9. Prove that number of subsets of a set containing n distinct elements is 2^n, for all $n \in N$.

10. If $S_n = 1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{n}$, $n \in N$, then prove that $S_1 + S_2 + S_3 + \ldots S_n = (n + 1)S_n - n$.

Ananth Garg

Ananth Garg

CHAPTER 4

DPP 4.1

Subjective Type Questions

1. Let $P(n)$: $1 + 5 + 9 + \ldots + (4n - 3) = n(2n - 1)$, for all natural numbers n.

Then $P(1) : 1 = 1(2 \times 1 - 1) = 1$, which is true.

Hence, $P(1)$ is true.

Let us assume that $P(n)$ is true for some natural number $n = k$.

$\therefore \quad P(k) : 1 + 5 + 9 + \ldots + (4k - 3) = k(2k - 1) \quad \ldots(i)$

Now, we have to prove that $P(k + 1)$ is also true.

$$P(k+1) : 1 + 5 + 9 + \ldots + (4k - 3) + [4(k + 1) - 3]$$
$$= 2k^2 - k + 4k + 4 - 3 \quad \text{(Using (i))}$$
$$= 2k^2 - k + 4k + 4 - 3 = 2k^2 + 3k + 1$$
$$= (k + 1)(2k + 1) = (k + 1)[2(k + 1) - 1]$$

Hence, $P(k + 1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for every natural number n.

2. Let

$$P(n): \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \ldots + \frac{2^n}{1+x^{2^n}} = \frac{1}{x-1} + \frac{2^{n+1}}{1-x^{2^{n+1}}},$$

where $x \neq \pm 1$ and $n \in N$.

When $n = 1$,

$$\text{L.H.S.} = \frac{1}{1+x} + \frac{2}{1+x^2} = \frac{x^2+2x+3}{(1+x)(x^2+1)}$$

$$\text{R.H.S.} = \frac{1}{x-1} + \frac{4}{1-x^4} = \frac{1}{x-1} - \frac{4}{(x-1)(x+1)(x^2+1)}$$

$$= \frac{x^3+x^2+x-3}{(x-1)(x+1)(x^2+1)} = \frac{x^2+2x+3}{(x+1)(x^2+1)}$$

Since L.H.S. = R.H.S., P(1) is true.

Let $P(n)$ be true for $n = k$. Then

$$\frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \ldots + \frac{2^k}{1+x^{2^k}} = \frac{1}{x-1} + \frac{2^{k+1}}{1-x^{2^{k+1}}}$$

For $n = k + 1$,

$$\frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \ldots + \frac{2^k}{1+x^{2^k}} + \frac{2^{k+1}}{1+x^{2^{k+1}}}$$

$$= \frac{2^{k+1}}{1+x^{2^{k+1}}} + \frac{1}{x-1} + \frac{2^{k+1}}{1-x^{2^{k+1}}}$$

$$= \frac{1}{x-1} + \frac{2 \times 2^{k+1}}{(1+x^{2^{k+1}})(1-x^{2^{k+1}})}$$

$$= \frac{1}{x-1} + \frac{2^{(k+1)+1}}{1-x^{2^{(k+1)+1}}}$$

So, $P(n)$ is true for $n = k + 1$ also.

Hence, $P(n)$ is true for all $n \in N$.

3. Let $P(n)$: $\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\ldots\left(1-\frac{1}{(n+1)^2}\right) = \frac{n+2}{2n+2}$, $n \in N$.

$$P(1) : \text{L.H.S.} = 1 - \frac{1}{2^2} = \frac{3}{4}$$

$$\text{R.H.S.} = \frac{1+2}{2+2} = \frac{3}{4}$$

So, $P(n)$ is true for $n = 1$.

Let $P(n)$ be true for $n = k$. Then

$$\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\ldots\left(1-\frac{1}{(k+1)^2}\right) = \frac{k+2}{2k+2}$$

For $n = k + 1$,

$$\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\ldots\left(1-\frac{1}{(k+1)^2}\right)\left(1-\frac{1}{(k+2)^2}\right)$$

$$= \frac{k+2}{2k+2}\left(\frac{(k+2)^2-1}{(k+2)^2}\right)$$

$$= \frac{1}{2k+2}\left(\frac{(k+3)(k+1)}{k+2}\right) = \frac{(k+1)+2}{2(k+1)+2}$$

So, $P(n)$ is true for $n = k + 1$ also.

Hence, $P(n)$ is true for all $n \in N$.

4. Let $P(n) : 2^{3n} - 1$ is divisible by 7

Now, $P(1) : 2^3 - 1 = 7$; This is divisible by 7.

Hence, $P(1)$ is true.

Assume that, $P(n)$ is true for some natural number $n = k$.

$\therefore \quad P(k) : 2^{3k} - 1$ is divisible by 7.

or $\quad 2^{3k} - 1 = 7m, m \in N \quad \ldots(i)$

Now, to prove $P(k + 1)$ is true.

$$P(k+1) : 2^{3(k+1)} - 1 = 2^{3k} \times 2^3 - 1$$
$$= 8(7m + 1) - 1 = 56m + 7$$
$$= 7(8m + 1), \text{ which is divisible by 7.}$$

Thus, $P(k + 1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for all natural numbers n.

5. Let $P(n) : 7^n - 2^n$ is divisible by 5, for any natural number n.

Now, $P(1) = 7^1 - 2^1 = 5$, which is divisible by 5.

Hence, $P(1)$ is true.

Assume that, $P(n)$ is true for some natural number $n = k$.

$\therefore \quad P(k) = 7^k - 2^k$ is divisibly by 5

or $\quad 7^k - 2^k = 5m, m \in N \quad \ldots(i)$

Now, let us prove that $P(k + 1)$ is true.

$$P(k+1): 7^{k+1} - 2^{k+1} = 7^k \cdot 7 - 2^k \cdot 2$$
$$= (5 + 2)7^k - 2^k \cdot 2 = 5 \cdot 7^k + 2 \cdot 7^k - 2 \cdot 2^k$$
$$= 5 \cdot 7^k + 2(7^k - 2^k) = 5 \cdot 7^k + 2(5m) \quad \text{(using (i))}$$
$$= 5(7^k + 2m), \text{ which is divisible by 5}$$

Thus, $P(k + 1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for all natural numbers n.

6. Let $P(n)$: $n^2 < 2^n$ for all natural numbers $n \geq 5$.

Now, $P(5)$: $5^2 < 2^5$ or $25 < 32$, which is true.

Hence, $P(5)$ is true.

Assume that, $P(n)$ is true for some natural number $n = k$.

$\therefore \quad P(k)$: $k^2 < 2^k$ ...(i)

Now, let us prove that $P(k + 1)$ is true.

$P(k + 1)$: $(k + 1)^2 < 2^{k+1}$

$(k + 1)^2 = k^2 + 2k + 1 < 2^k + 2k + 1$ (using (i)) ...(ii)

Now let, $2^k + 2k + 1 < 2^{k+1}$ ...(iii)

$\therefore \quad 2^k + 2k + 1 < 2 \cdot 2^k$

$\Rightarrow \quad 2k + 1 < 2^k$, which is true for all $k \geq 5$

Thus, using (ii) and (iii), $(k + 1)^2 < 2^{k+1}$

Hence, $P(k + 1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number $n \geq 5$.

7. We have a sequence $d_1, d_2, d_3, \ldots$ is defined by letting $d_1 = 2$ and $d_k = \dfrac{d_{k-1}}{k}$.

Let $P(n) : d_n = \dfrac{2}{n!} \ \forall\, n \in N$

$$P(2) : d_2 = \frac{2}{2!} = \frac{2}{2 \times 1} = 1$$

Also, $d_1 = 2$ and $d_k = \dfrac{d_{k-1}}{k}$

$$\Rightarrow \quad d_2 = \frac{d_1}{2} = \frac{2}{2} = 1$$

Hence, $P(2)$ is true.

Now, assume that $P(n)$ is true for $n = k$.

$\therefore \quad P(k) : d_k = \dfrac{2}{k!}$ ...(i)

To prove that $P(k + 1)$ is true, we have to show that $P(k + 1)$:

$$d_{k+1} = \frac{2}{(k+1)!}.$$

Now, $d_{k+1} = \dfrac{d_{k+1-1}}{k} = \dfrac{d_k}{k} = \dfrac{2}{k!k} = \dfrac{2}{(k+1)!}$

Hence, $P(k + 1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number n.

8. Let $P(n)$: $\dfrac{1}{n+1} + \dfrac{1}{n+2} + \ldots + \dfrac{1}{2n} > \dfrac{13}{24}$, for all natural numbers $n > 1$.

$$P(2) : \frac{1}{2+1} + \frac{1}{2+2} > \frac{13}{24}$$

$$\Rightarrow \quad \frac{1}{3} + \frac{1}{4} > \frac{13}{24} \quad \Rightarrow \quad \frac{4+3}{12} > \frac{13}{24}$$

$$\Rightarrow \quad \frac{7}{12} > \frac{13}{24}, \text{ which is true.}$$

Hence, $P(2)$ is true.

Assume that $P(n)$ is true, for $n = k$.

$\therefore \quad P(k) : \dfrac{1}{k+1} + \dfrac{1}{k+2} + \ldots + \dfrac{1}{2k} > \dfrac{13}{24}$ ...(i)

Now, to prove $P(k + 1)$ is true, we have to show that

$$P(k+1): \frac{1}{k+1} + \frac{1}{k+2} + \ldots + \frac{1}{2k} + \frac{1}{2(k+1)} > \frac{13}{24}$$

$$\frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{2k} + \frac{1}{2(k+1)} > \frac{13}{24} + \frac{1}{2(k+1)} > \frac{13}{24}$$

$$\left(\because \frac{1}{2(k+1)} > 0\right)$$

Hence, $P(k + 1)$ is true whenever $P(k)$ is true.

So, by the principle of mathematical induction, $P(n)$ is true for any natural number $n > 1$.

9. Let $P(n)$: Number of subset of a set containing n distinct elements is 2^n, for all $n \in N$.

For $n = 1$, consider set $A = \{1\}$.

So, set of subsets is $\{\{1\}, \phi\}$, which contains 2^1 elements.

So, $P(1)$ is true.

Assume that $P(n)$ is true, for $n = k$.

$\therefore \quad P(k)$: Number of subsets of a set containing k distinct elements is 2^k.

To prove $P(k + 1)$ is true, we have to show that

$P(k + 1)$: Number of subsets of a set containing $(k + 1)$ distinct elements is 2^{k+1}.

We know that, with the addition of one element in the set, the number of subsets become double.

$\therefore$ Number of subsets of a set containing $(k + 1)$ distinct elements $= 2 \times 2^k = 2^{k+1}$

So, $P(k + 1)$ is true. Hence, $P(n)$ is true.

10. Let $P(n) : S_1 + S_2 + S_3 + \ldots S_n = (n + 1)S_n - n$, where

$$S_n = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}, \ n \in N.$$

For $P(1)$, $S_1 = 1$ and $(1 + 1)S_1 - 1 = 1$

So, $P(n)$ is true for $n = 1$.

Let $P(n)$ is true $n = k$.

$\therefore \quad P(k) : S_1 + S_2 + S_3 + \ldots S_k = (k + 1)S_k - k$, where

$$S_k = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{k}.$$

For $n = k + 1$, $S_1 + S_2 + S_3 + \ldots S_k + S_{k+1}$

$$= (k+1)S_k - k + 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{k} + \frac{1}{k+1}$$

$$= (k+1)S_k - k + S_k + \frac{1}{k+1}$$

$$= (k+2)S_k - k + \frac{1}{k+1} \quad \ldots(1)$$

and $(k + 2)S_{k+1} - (k + 1)$

$$= (k+2)\left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{k} + \frac{1}{k+1}\right) - k - 1$$

$$= (k+2)S_k + \frac{k+2}{k+1} - k - 1$$

$$= (k+2)S_k - k + \frac{1}{k+1} \quad \ldots(2)$$

So, from (1) and (2), $P(n)$ is true for $n = k + 1$.

Hence, $P(n)$ is true for all $n \in N$.